

Ring

subring.

i) $(S, +) \leq (R, +) \Leftrightarrow \forall a, b \in S \quad a+b \in S$

ii) $\forall a, b \in S \quad ab \in S$

$N: (R, +)$ has identity.

$(S, +, \cdot)$ may be has no identity.

Example: $M_n(\mathbb{Z})$ has a subring $M_n(2\mathbb{Z})$

S is a subring of R .

$1_S \neq 1_R$

Example $M_n(\mathbb{Z})$ $n \geq 2$ $S = \left\{ \begin{pmatrix} A & 0 \\ 0 & 0 \end{pmatrix} \mid A \in M_{n-1}(\mathbb{Z}) \right\}$
 E_n $\text{diag}(E_{n-1}, 0)$

$X \subseteq R$

$\langle X \rangle = \bigcap \{ S \in R \}$

$= \langle x_1^{n_1} \dots x_r^{n_r} \mid x_i \in S, n_i \in \mathbb{N} \rangle$

$\hookrightarrow$ subgroup of $(R, +)$

$\left\{ \sum_{\text{finite sum}} a_{n_1 \dots n_r} x_1^{n_1} \dots x_r^{n_r} \mid x_i \in X, n_i \in \mathbb{N}, r \in \mathbb{N}, a_{n_1 \dots n_r} \in \mathbb{Z} \right\}$

finite ring. $(R, +, \cdot)$

$(R, +, \cdot) \cong \mathbb{Z}_{p^e} \oplus \dots \oplus \mathbb{Z}_{p^e}$

$|R| < \infty$ has no nonzero divisor

$|R| \geq 2$ is a field.

$\Rightarrow |R| = p^n \quad x^p - x = 0$

R is the set of all roots of $x^p - x = 0$

$\phi: I \rightarrow R'$ mapping

① $\phi(a+b) = \phi(a) + \phi(b)$

② $\phi(ab) = \phi(a)\phi(b)$

ϕ is a homomorphism.

ϕ : monomorphism

epimorphism

homomorphism

同态基本定理

i) $\text{Ker } \phi = \{ a \in R \mid \phi(a) = 0 \} \leq (R, +)$

$\forall a \in \text{Ker } \phi, \forall b \in R, \quad ab \in \text{Ker } \phi \Rightarrow \text{Ker } \phi$ is a subring of R

Ex: I is a subring of R

① $\forall a \in R, \forall x \in I, \quad ax \in I \Rightarrow I$ is a left ideal.

② $\forall a \in R, \forall x \in I, \quad xa \in I \Rightarrow I$ is a right ideal

③ $\forall x \in I, \forall a \in R, \quad \frac{xa}{ax} \in I \Rightarrow I$ is an ideal of R

$\text{Im } \phi = \{ \phi(a) \mid a \in R \} \leq R$

Property. ① ϕ is injective $\Leftrightarrow \text{Ker } \phi = \{ 0 \}$ zero ideal.

② ϕ is surjective $\Leftrightarrow \text{Im } \phi = R'$

Prop. Suppose I is an ideal of $R (I \trianglelefteq R)$

$(R/I, +)$ is an abelian group $\Rightarrow$ becomes a

ring with product. $= \{ a+I \mid a \in R \}$

$(a+I)(b+I) := ab+I$

Well-defined $a+I = a'+I \quad b+I = b'+I$

Pf: $a' = a+x \quad b' = b+y \quad x, y \in I$

$a'b'+I = ab+xb+ay+xy+I = ab+I$

$(\mathbb{Z}_n, +, \cdot) = \{ \bar{a} \mid 0 \leq a \leq n-1 \}$

$\parallel$
 $a+n\mathbb{Z}$

$I = n\mathbb{Z}$ is an ideal $\mathbb{Z} \quad nk - nl = n(k-l) \in I$

$\forall nk \in I$

$m \cdot nk = n(mk) \in n\mathbb{Z}$

$\forall m \in \mathbb{Z}$

$\mathbb{Z}/n\mathbb{Z} = \{ a+n\mathbb{Z} \mid a \in \mathbb{Z} \}$

$(a+n\mathbb{Z}) + (b+n\mathbb{Z}) = (a+b)+n\mathbb{Z}$

$\bar{a} + \bar{b} = \overline{a+b}$

$(a+n\mathbb{Z})(b+n\mathbb{Z}) = ab+n\mathbb{Z}$

Example $\mathbb{R}[x]$ $I = \{ x^2 + f(x) \mid f(x) \in \mathbb{R}[x] \}$
 $= (x^2)$

$\mathbb{R}[x]/I = \{ f(x)+I \mid f(x) \in \mathbb{R}[x] \}$

$= \{ ax+b + \underbrace{g(x)x^2+I}_{g(x)x^2 \in I} \mid a, b \in \mathbb{R}, g(x) \in \mathbb{R}[x] \}$

$= \{ ax+b+I \mid a, b \in \mathbb{R} \}$

$= \mathbb{R}[x]/I \cong \mathbb{R}(x)$
as vector space

$\{1+I, x+I\}$ is a basis of $R[x]/I$

i) $a+bx+I = a(1+I)+b(x+I)$

ii) $a(1+I)+b(x+I) = a+bx+I = 0$

$\Rightarrow a=b=0$. $a+bx+I=0 \Rightarrow a+bx \in I$

$a+bx \in \{a^2+b^2\}f(x)$. $f(x) \in R[x]$

in $R[x]$. $\Rightarrow f(x)=0$

$\Rightarrow a=b=0$

$(x+I)^2 = x^2+I = -1+I$

$R[x]/(x^2+1) \rightarrow \mathbb{C}$

$a+bx+I \rightarrow a+bi$

First. $\varphi: R \rightarrow S$ homomorphisms.

$\bar{\varphi}: R/\ker \varphi \rightarrow \text{Im } \varphi$

$a + \ker \varphi \mapsto \varphi(a)$

Pf. 1st. hom of groups $\Rightarrow \bar{\varphi}$ iso of groups

$\bar{\varphi}((a+\ker \varphi)(b+\ker \varphi))$

$= \bar{\varphi}(ab + \ker \varphi) = \varphi(ab)$

$= \varphi(a)\varphi(b) = \bar{\varphi}(a+\ker \varphi)\bar{\varphi}(b+\ker \varphi)$

2nd. $\varphi: R \rightarrow S$ hom.

i) $T \leq R \Rightarrow \varphi(T) \leq S$

ii) $S' \leq S \Rightarrow \varphi^{-1}(S') = \{a \in R \mid \varphi(a) \in S'\} \leq R$
and $\varphi^{-1}(S') \supseteq \ker \varphi$

iii) $J \triangleleft S \xrightarrow{\varphi^{-1}} \varphi^{-1}(J) \xrightarrow{\varphi} J$ is an ideal of R .

iv) $\varphi: R \rightarrow S$ epic, $I \triangleleft R \Rightarrow \varphi(I) \triangleleft S$

App $\pi: R \rightarrow R/I$ canonical map

$x \mapsto x+I$

$\Rightarrow I \triangleleft R/I \Rightarrow I = J/I (J \supseteq I, J \triangleleft R)$

3rd. $I \triangleleft R, J \leq R$

① $I+J \leq R$

② $I \triangleleft I+J$

③ $I \cap J \triangleleft J$

④ $I+J/I \cong J/I \cap J$

$\forall a_i + b_i (i=1,2) \in I+J, a_i \in I, b_i \in J$

$(a_1+b_1)-(a_2+b_2) = (a_1-a_2)+(b_1-b_2) \in I+J$

$(a_1+b_1)(a_2+b_2) = \underbrace{a_1a_2}_I + \underbrace{a_1b_2 + b_1a_2}_J + \underbrace{b_1b_2}_J$

Example $M_2(\mathbb{R})$ $S = \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} \mid a, b \in \mathbb{R} \right\}$

$S_2 = \left\{ \begin{pmatrix} 0 & 0 \\ x & x \end{pmatrix} \mid x \in \mathbb{R} \right\}$

$S_1 + S_2 = \left\{ \begin{pmatrix} a & b \\ x & a+x \end{pmatrix} \mid a, b, x \in \mathbb{R} \right\}$

$\begin{pmatrix} a & b \\ x & a+x \end{pmatrix} \begin{pmatrix} c & d \\ y & c+y \end{pmatrix} = \begin{pmatrix} ac+by & ad+bc+by \\ cx+ay & dx+ac+ay+xc \end{pmatrix}$

不封闭: $S_1 + S_2$ 不一定为子环

$\langle x \rangle$

$\langle \mathbb{Z} \rangle = \{ a_0 + a_1 \mathbb{Z} + \dots + a_n \mathbb{Z}^n \mid a_i \in \mathbb{Z}, n \in \mathbb{Z} \}$

$= \mathbb{Z}[\mathbb{Z}]$

Example $R = M_2(\mathbb{R})$ $\mathbb{Z} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$

$\langle \mathbb{Z} \rangle = \{ a_0 + a_1 \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} + \dots + a_n \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n \mid a_i \in \mathbb{Z} \}$

$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$

$= \left\{ \begin{pmatrix} a & b \\ 0 & a \end{pmatrix} \mid a, b \in \mathbb{Z} \right\}$

$\langle x \rangle = \{ nx + rx \mid n \in \mathbb{Z}, r \in R \}$

is a right ideal containing x

① suppose J is a right ideal of $R, x \in J$
Then $J \supseteq \langle x \rangle$

Pf ① $nx + rx, nx + rx_2 \in \langle x \rangle$

$(n_1x + r_1x)(n_2x + r_2x) = (n_1n_2)x + x(r_1r_2) \in \langle x \rangle$

i.e. $\langle x, + \rangle \in (R, +)$

$\forall y \in R, (n_1x + r_1x)y = 0 \cdot x + x(n_1y + r_1y) \in \langle x \rangle$

$\therefore \langle x \rangle$ 为右理想

② $\forall nx \in J (J \leq (R, +))$ $x \in J$ (J right ideal)

$nx + xr \in J \Rightarrow \langle x \rangle \subseteq J$

$\langle x \rangle = \{ nx + rx \mid r \in R, n \in \mathbb{Z} \}$

$\langle x \rangle = \left\{ nx + rx + xS + \sum_{i=1}^k r_i x S_i \mid n \in \mathbb{Z}, k \in \mathbb{N}, r, S, r_i, S_i \in R \right\}$

① $\langle x \rangle \triangleleft R$ containing x .

$J \triangleleft R$ and $x \in J \Rightarrow \langle x \rangle \subseteq J$

Def principle ^{right} ideal
every ideal of $\mathbb{Z}_n$ is principal.

Pf: $I \trianglelefteq \mathbb{Z}_n$. Since I is a subgroup
of $\mathbb{Z}/n\mathbb{Z}$: $\exists m$ s.t. $I = \frac{m\mathbb{Z} + n\mathbb{Z}}{n\mathbb{Z}}$

$$I = m\mathbb{Z}/n\mathbb{Z} = (m)$$

In a commutative ring.

$$(x) = \{ nx + rx \mid r \in R, n \in \mathbb{Z} \}$$

with identity: $xR = (x)$